

“Keil uVision Project Creation

This document gives instructions on how to create a keil project from scratch with nothing

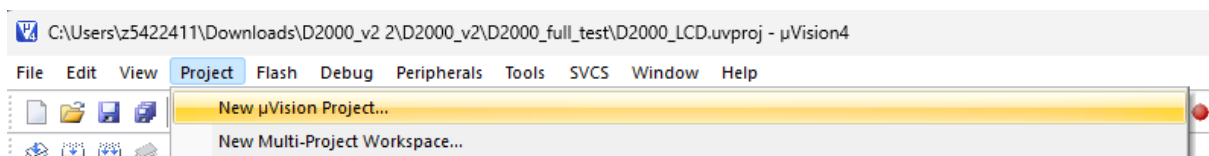
I have located useful headers within "D2000_Empty/headers", these are CMSIS headers and stm32 registers.

They are useful for styling and quality of life, but you don't really need them.

Step 1 and Step 2 are done for you, but here just so you know how you would do it.
Step 3 is needed, please read it

Step 1 – Project Creation

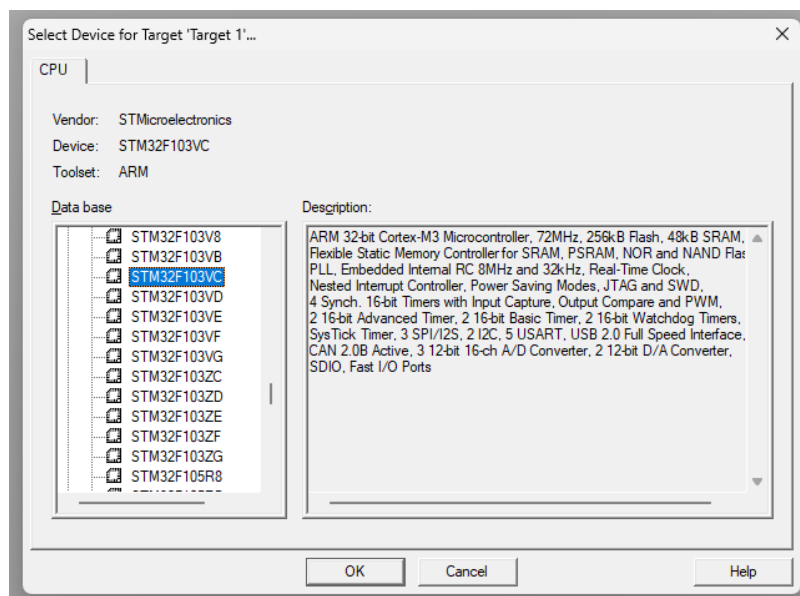
open keil uVision 4 and on the top panel, click on Project->new uVision Project



Then select a folder for you to put your project in. As for this empty project template, place it in “D2000 Empty/KeiluVision4”.

Give it a name, for example “joeiscool” and save it.

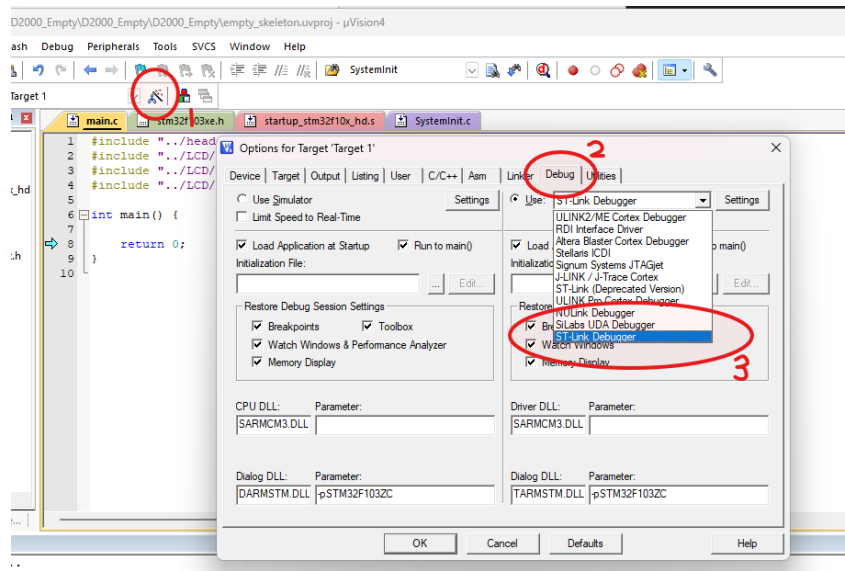
After that you will be prompted to select a device for target. If you're using APE Board, you should select STMicroelectronics->STM32F103ZC.



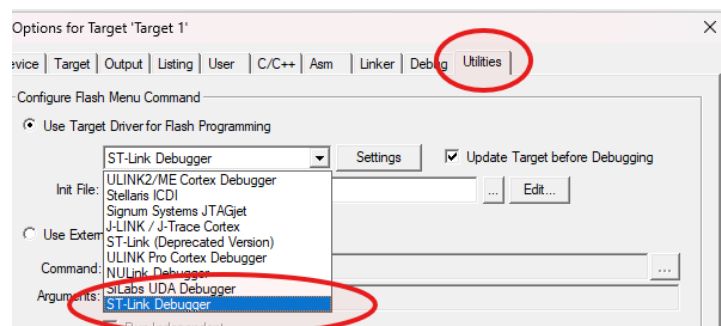
This just comes down to which microcontroller you are using. Click OK.

Then you will be prompted Copy 'startup_stm32f10x_hd.s' to Project Folder and add file to project. Hit yes.

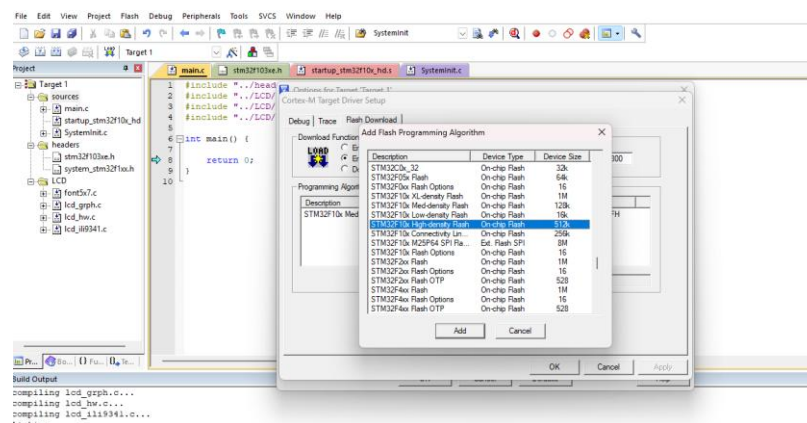
Your project has been created! Add this point, you might need to go to target options and change your debugger to an ST link



Then go to utilities and use STlink debugger for flash programmer



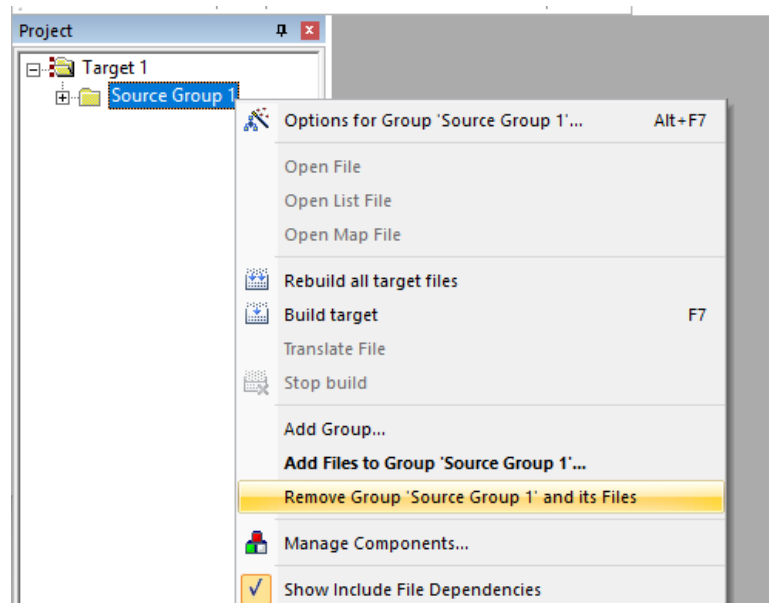
Then click settings and add STM32F10x High-density Flash



Step 2 – File and Folder Management

The 'startup_stm32f10x_hd.s' is the startup file is now located in "*D2000_Empty*". But that is not very nice (ugly), so go to your file system and move this file to "*D2000_Empty/sources*".

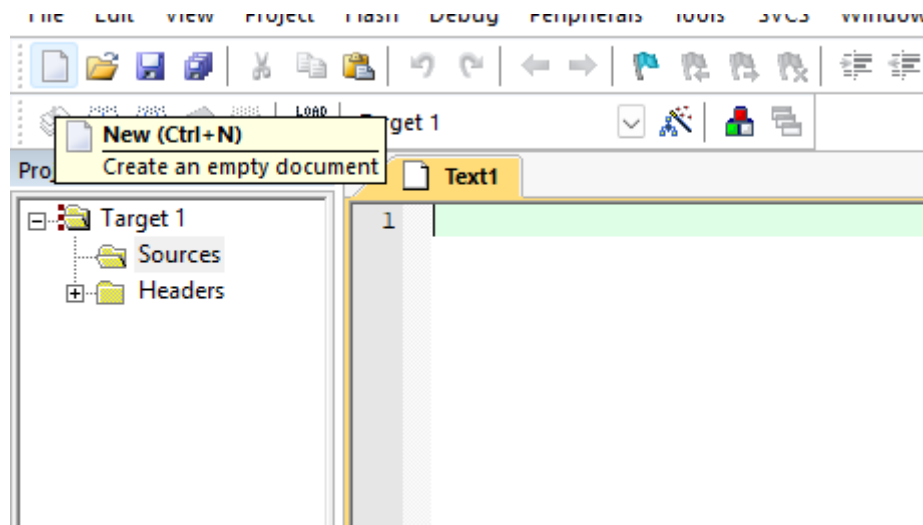
In Keil, remove the "*Source Group 1*" because the name is ugly



And now, right click on "Target 1", the "Add Group", and finally name this new group "*Sources*", do the same for "*Headers*".

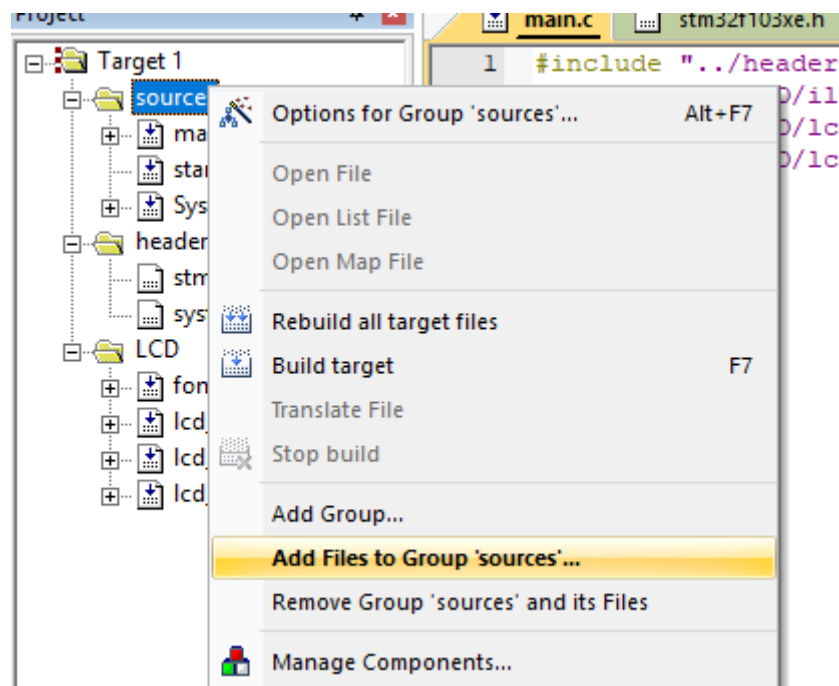
Step 3 – Adding File

[Click here to create a new file](#)



Then ctrl + S to save, navigate to “../source” and name the file *filename.c*.

Then right click on *sources* , then click “Add Files to Group “sources” here and find the *filename.c* file you just referenced, then find your *filename.c* file



The rest is just C, which won't be explained in the document. For instance, function prototype, header files etc. But the LCD library has been included for your convenience. Have fun.